

1. Product Overview

WIN-SN-PM-M is a compact, high-precision Particulate matter (PM1.0, PM2.5, and PM10) sensor designed for applications requiring low power consumption and small form factor. It ensures reliable measurements over a wide range. Its RS485 communication interface simplifies integration into small, embedded systems, making it ideal for IoT, HVAC, buildings, and for environmental quality monitoring. The sensor's long-term stability ensures consistent performance in harsh environments (Wind/Birds), making it a cost-effective solution for long-term deployment. Additionally, it features a 4-digit 7-segment display that provides real-time measurement data, allowing users to monitor sensor readings directly without requiring an external interface.



Fig.1 - Product Image

- Particulate matter sensor with range of 0-1000ug/m3
- Superior sensor performance.
- Operating range -10°C to 60°C for temperature and 0% to 80% RH for humidity
- Fully calibrated and processed digital RS485 Output
- Excellent long-term stability.
- In built 7 segment display for real time data of parameters.

Sensor specification

Measuring Parameter	Measuring Range	Accuracy
PM 1.0	0 ~ 1000 ug/m3	5 µg/m3 10 % m.v.
PM 2.5	0 ~ 1000 ug/m3	5 µg/m3 10 % m.v.
PM 10	0 ~ 1000 ug/m3	25 µg/m3 25 % m.v.

	Product Configuration
☐	PM1/2.5/10+M
☐	PM1/2.5/10+M+D

2. Sensor Specifications

- **Wide temperature range:** -40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 100*100*60 mm.
- **Certifications:**



3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

4. Configuration Settings

- Communication Speed 9600 – 115200 Kbps (SW Selectable)
- Data Bits 8
- Parity None
- CRC Yes
- Slave ID 1, DIP (SW Selectable)
- Function Code 0X03 (Read Holding Register)

Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

5. Modbus RS485 Data Storage register

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	PM 1.0	40001	0x03	RS485	16-bit int
2	PM 2.5	40002	0x03	RS485	16-bit int
3	PM 10	40003	0x03	RS485	16-bit int
1	Display Baud Rate (Default: 960)	40011	0x03	RS485	16-bit int
2	Enter New Baud Rate	40012	0x03	RS485	16-bit int
3	Display Slave ID (Default: 1)	40013	0x03	RS485	16-bit int
4	Enter New Slave ID	40014	0x03	RS485	16-bit int
5	Display Parity (Default: None-3 And for Odd-1, Even-2)	40015	0x03	RS485	16-bit int
6	Enter New Parity	40016	0x03	RS485	16-bit int
7	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16-bit int
8	Enter New Start/ Stop Bit	40018	0x03	RS485	16-bit int

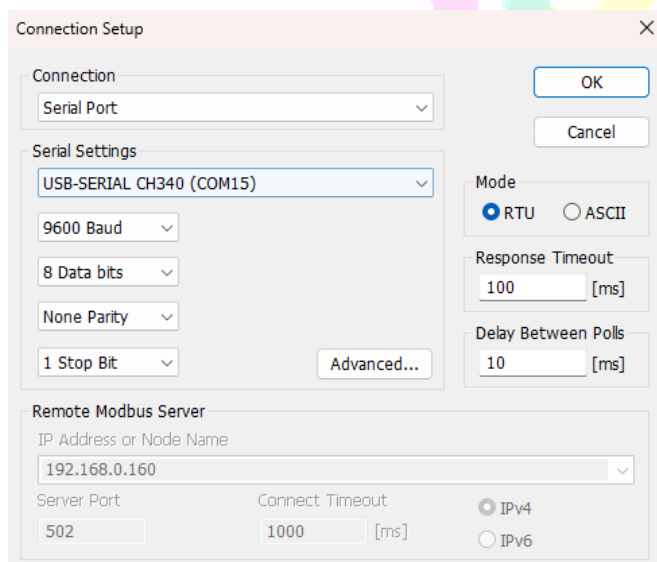
6. Modbus Configuration

Here is an example of receiving data on Mb poll software.

- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below.

Important Note

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc, enter it like this 960/1920/3840/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate (115200) and 2 at 40013 slave id.



Connection Setup

Connection: Serial Port

Serial Settings: USB-SERIAL CH340 (COM15)

9600 Baud

8 Data bits

None Parity

1 Stop Bit

Advanced...

Mode: ☒ RTU ☐ ASCII

Response Timeout: 100 [ms]

Delay Between Polls: 10 [ms]

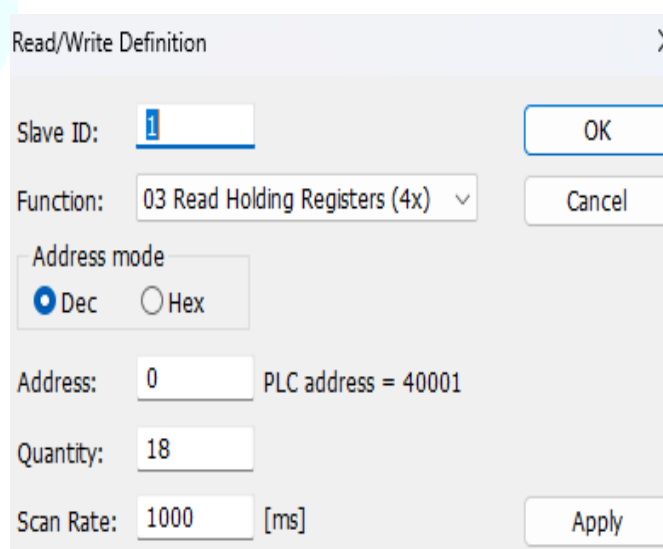
Remote Modbus Server

IP Address or Node Name: 192.168.0.160

Server Port: 502

Connect Timeout: 1000 [ms]

☒ IPv4 ☐ IPv6



Read/Write Definition

Slave ID: 1

Function: 03 Read Holding Registers (4x)

Address mode: ☒ Dec ☐ Hex

Address: 0 PLC address = 40001

Quantity: 18

Scan Rate: 1000 [ms]

Apply

Mbpoll1

Tx = 2844: Err = 51: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	PM 1.0	912	Current Baud	960
1	PM 2.5	884	New Baud	0
2	PM 10	108	Current SID	1
3		0	New SID	0
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

- Above image Shows sensor data over Modbus with default Modbus settings
- At the register 40001 shows PM1.0 Values, register 40002 shows PM2.5 Values, register 40003 shows PM10 Values.
- As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id Jumper should be in single/removed then, you must enter a value on the 40012 resistors for the baud rate and 40014 for the slave id as shown in the image below and restart the board.
- Now you can see data below image set 115200 baud rate and slave ID 10.

Mbpoll1

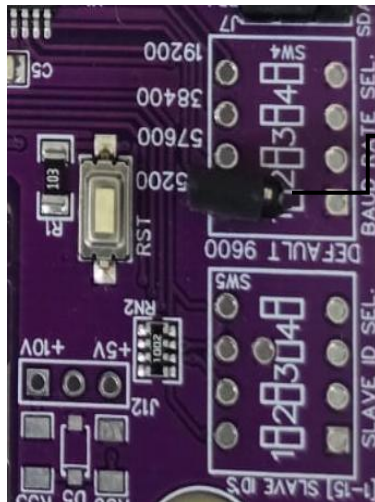
Tx = 2844: Err = 51: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	PM 1.0	912	Current Baud	960
1	PM 2.5	884	New Baud	11520
2	PM 10	108	Current SID	1
3		0	New SID	10
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

Mbpoll1

Tx = 2844: Err = 51: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0	PM 1.0	912	Current Baud	11520
1	PM 2.5	884	New Baud	0
2	PM 10	108	Current SID	10
3		0	New SID	0
4		0	Current Parity	3
5		0	New Parity	0
6		0	Current Stopbit	1
7		0	New Stopbit	0
8		0		0
9		0		0

**Image1.1****Image1.2**

Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the image 1.2) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

Contact Information

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